

Experiment :- 5

AIM :- Create and understand the Software Requirement Specification (SRS) document for the BizFlow Inventory and Billing Management System.

Introduction :-

Software Requirement Specification (SRS) is a document that describes all the requirements of a software system.

It acts as a blueprint for software development.

The SRS document helps developers, testers, and users understand:

- What the system should do
- How the system should perform
- Hardware and software requirements
- Functionalities of the project

Purpose of the Project :-

The purpose of the BizFlow Inventory and Billing Management System is:

- To manage product inventory efficiently
- To generate customer bills quickly
- To automate inventory updates
- To maintain sales records
- To generate invoices in PDF format
- To provide sales analytics and business insights

Scope of the Project :-

The system is designed for:

- Small businesses
- Retail stores
- Wholesale distributors
- Inventory management
- Billing and invoicing
- Sales tracking and analysis

The project helps business owners manage inventory, billing operations, and sales records through a centralized software platform.

Objectives :-

Main Objectives:

1. Manage product inventory
2. Generate customer bills
3. Track stock availability
4. Generate PDF invoices
5. Maintain bill history
6. Analyze sales performance
7. Improve business efficiency

Overall Description :-

The system allows users to manage products, monitor inventory, create bills, and analyze sales data.

The system:

- Stores product information
- Tracks inventory levels
- Generates bills and invoices
- Maintains sales history
- Updates inventory automatically
- Displays analytics through a dashboard

Functional Requirements

Functional Requirements describe what the system should do.

i) User Login Module

- Admin can log into the system.
- Username and password validation is required.

ii) Product Management Module

- User can add products.
- User can update product details.

- User can delete products.
- Product information should be stored in the database.

iii) Inventory Management Module

- System should maintain product stock levels.
- Inventory should update automatically after billing.
- Stock availability should be displayed.

iv) Billing Module

- User should be able to create bills.
- Multiple products can be added to a bill.
- Bill amount should be calculated automatically.

v) Invoice Generation Module

- System should generate professional PDF invoices.
- Invoice should contain product and billing details.

vi) Bill History Module

- System should store all generated bills.
- Previous bills should be searchable and viewable.

vii) Analytics Dashboard Module

- System should display sales statistics.
- Revenue and sales trends should be shown.
- Business performance reports should be available.

Non-Functional Requirements

Non-functional requirements describe system performance and quality.

i) Performance Requirement

- System should respond quickly to user actions.
- Billing operations should be processed efficiently.

ii) Security Requirement

- User authentication should be secure.
- Business data should be protected from unauthorized access.

iii) Reliability Requirement

- System should operate continuously with minimum failures.
- Data consistency should be maintained.

iv) Scalability Requirement

- New products and sales records can be added without affecting performance.
- Additional features can be integrated in future versions.

v) Usability Requirement

- User interface should be simple and easy to use.
- Navigation should be user-friendly.

Hardware Requirements

- Computer/Laptop
- Minimum 4 GB RAM
- Internet Connection
- Storage Device

Software Requirements

- Operating System (Windows/Linux)
- Web Browser
- React.js
- Vite
- Supabase
- PostgreSQL
- JavaScript

Expected Outcomes

- Efficient inventory management
- Faster billing process
- Accurate sales record maintenance
- Automated invoice generation
- Better business decision making through analytics
- Reduced manual work and human errors